

B5 NegotiationArena Cross-Lingual Contact: Seed-101 OpenAI Evidence

negotiation_arena_crosslingual

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Answer

For the current B5 NegotiationArena evidence, the answer is a constrained null: the seed-101 OpenAI `gpt-5.4-mini-2026-03-17` runs do not support a stable higher-resource language payoff premium after counterbalancing. All buy/sell C2 rows reached deals and parsed offers, but the payoff differences were role-sensitive and two C2 rows failed strict assigned-channel compliance because visible dialogue converged to Chinese. These are OpenAI benchmark override results, not Qwen3 evidence.

H1 is not supported. EN-ID showed EN minus ID payoff asymmetry of +36 when EN was buyer, but 0 when EN was seller. EN-ZH flipped sign, -10 then +8. ZH-ID was -20 then 0 for ZH minus ID. H2 therefore fails as a language-premium claim: the counterbalances do not show a stable same-sign premium. H3 is weak and mostly first-offer/seller driven rather than higher-resource-channel driven. Final prices were 0–10 ZUP away from the first offer, and the first offer was always made by the seller. H4 is supported only as a convergence observation: C3 converged to EN for EN-ID, to ZH for EN-ZH, and to ZH for ZH-ID, with no detected off-pair English leakage in ZH-ID. H5 is the best current summary: the capability floor passed, C2 outcomes were not worse than baselines, and no counterbalanced language premium remains defensible from this evidence.

Design and Gates

Language is the visible interaction-output channel. Rules and private state remain in English. The active language pairs are EN-ID, EN-ZH, and ZH-ID. C0 and C1 are monolingual output-channel baselines; C2 assigns each role one language and counterbalances buyer/seller role; C3 permits either language in the active pair. Source bring-up used the local NegotiationArena checkout at `external/NegotiationArena`, branch `paper_experiment_code`, commit `d35a7a3aa0d94c2d49f1d6ac13c5f931851abf12`.

The G2 capability gate passed. C0 EN buy/sell reached a deal with offer parse rate 1.0 and channel compliance 1.0 at `artifacts/results/baseline_c0_buy_sell_en_seed001.openai_benchmark.metrics.json`. C1 ID buy/sell likewise reached a deal with offer parse rate 1.0 and channel compliance 1.0 at `artifacts/results/baseline_c1_buy_sell_id_seed001.metrics.json`. C1 ZH buy/sell reached a deal with offer parse rate 1.0 and channel compliance 1.0 at `artifacts/results/pair_en_zh_c1_buy_sell_seed101.metrics.json`. The gate artifact is `artifacts/results/g2_capability_floor.json`.

Evidence Table

Pair/condition	Assignment	Final	Payoffs	Lx-Ly payoff	Channel compliance	Interpretation
EN-ID C2	buyer EN, seller ID	price 52	buyer 48, seller 12	+36	1.00	EN advantage only with EN as buyer
EN-ID C2	buyer ID, seller EN	price 70	buyer 30, seller 30	0	1.00	counterbalance removes EN premium
EN-ZH C2	buyer EN, seller ZH	price 75	buyer 25, seller 35	-10	0.50	Chinese convergence; no EN premium
EN-ZH C2	buyer ZH, seller EN	price 74	buyer 26, seller 34	+8	0.667	sign flip; compliance limited
ZH-ID C2	buyer ZH, seller ID	price 80	buyer 20, seller 40	-20	1.00	no ZH premium
ZH-ID C2	buyer ID, seller ZH	price 70	buyer 30, seller 30	0	0.50	Chinese convergence; no stable premium
EN-ID C3	bilingual EN/ID	price 50	buyer 50, seller 10	n/a	1.00	converged to EN; off-pair 0
EN-ZH C3	bilingual EN/ZH	price 50	buyer 50, seller 10	n/a	1.00	converged to ZH; off-pair 0
ZH-ID C3	bilingual ZH/ID	price 60	buyer 40, seller 20	n/a	1.00	converged to ZH; EN leakage 0

The C2 buy/sell rows cite, in order: `artifacts/results/pair_en_id_c2_buyer_lx_seller_ly_buy_sell_seed101.metrics.json`, `artifacts/results/pair_en_id_c2_buyer_ly_seller_lx_buy_sell_seed101.metrics.json`, `artifacts/results/pair_en_zh_c2_buyer_lx_seller_ly_buy_sell_seed101.metrics.json`, `artifacts/results/pair_en_zh_c2_buyer_ly_seller_lx_buy_sell_seed101.metrics.json`, `artifacts/results/pair_zh_id_c2_buyer_lx_seller_ly_buy_sell_seed101.metrics.json`, and `artifacts/results/pair_zh_id_c2_buyer_ly_seller_lx_buy_sell_seed101.metrics.json`. The C3 rows cite `artifacts/results/pair_en_id_c3_buy_sell_seed101.metrics.json`, `artifacts/results/pair_en_zh_c3_buy_sell_seed101.metrics.json`, and `artifacts/results/pair_zh_id_c3_buy_sell_seed101.metrics.json`.

First-Offer Anchoring

The buy/sell first-offer metric is final price minus first offer price. The EN-ID rows were -8 when the ID seller opened at 60 and -10 when the EN seller opened at 80. EN-ZH was -5 when the ZH seller opened at 80 and -6 when the EN seller opened at 80. ZH-ID was 0 when the ID seller opened at 80 and -10 when the ZH seller opened at 80. This is not a clean higher-resource anchoring effect; it is more consistent with seller-first-offer anchoring plus role and compliance effects. The one completed resource-exchange C2 row, `artifacts/results/pair_en_id_c2_buyer_lx_seller_ly_resource_exchange_seed101.metrics.json`, ended exactly at the EN agent’s first allocation offer, but resource exchange is not yet counterbalanced.

Representative Interaction Excerpts

EN-ID C2 buyer-EN/seller-ID: the ID seller opened, “Saya menawarkan X dengan harga 60 ZUP,” the EN buyer countered, “I can do 50 ZUP for X,” and the ID seller accepted 52 ZUP. Full transcript: `artifacts/transcripts/pair_en_id_c2_buyer_lx_seller_ly_buy_sell_seed101.json`.

EN-ZH C2 buyer-EN/seller-ZH: the seller opened in Chinese with a translated offer of 1 X for 80 ZUP, and the buyer assigned to English also replied in Chinese with a 70 ZUP counteroffer. This explains the 0.50 assigned-channel compliance despite a valid deal. Transcript: `artifacts/transcripts/pair_en_zh_c2_buyer_lx_seller_ly_buy_sell_seed101.json`.

ZH-ID C2 buyer-ID/seller-ZH: the ZH seller opened at 80 ZUP; the ID-assigned buyer replied in Chinese with 60 ZUP and later accepted 70 ZUP. This row is useful evidence of channel drift, not a clean assigned-language treatment. Transcript: `artifacts/transcripts/pair_zh_id_c2_buyer_ly_seller_lx_buy_sell_seed101.json`.

C3 free choice: EN-ID used English throughout at `artifacts/transcripts/pair_en_id_c3_buy_sell_seed101.json`; EN-ZH used Chinese throughout at `artifacts/transcripts/pair_en_zh_c3_buy_sell_seed101.json`; ZH-ID used Chinese throughout, with no detected English leakage, at `artifacts/transcripts/pair_zh_id_c3_buy_sell_seed101.json`.

Limitations and Next Work

This report is intentionally narrow. It is a seed-101 OpenAI benchmark evidence report and should not be generalized as a Qwen3 matrix result. The buy/sell matrix is counterbalanced for all active pairs, but it has only one seed per row. EN-ZH and one ZH-ID C2 row are compliance-limited because agents converged to Chinese despite assigned channels. Resource exchange is executable for all active pairs, but only the EN-ID agent-a-EN/agent-b-ID row has been run; the next exact resource-exchange commands are listed in `reports/findings.md`. More seeds and completed resource-exchange counterbalances are needed before estimating effect sizes.

Conclusion

The current answer to the headline question is no: in this evidence bundle, the higher-resource visible language channel does not capture a robust counterbalanced payoff premium. The clearest positive finding is process-level: free-choice language selection can converge strongly to one visible language, and in these seed-101 runs it converged to EN for EN-ID but to ZH for EN-ZH and ZH-ID.